

Experiment	<i>Lab 02: Experiment CYL – Flow Around Cylinders</i>
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Section Number	04 – Robert Bryan

Summary

This experiment's primary objective is to comprehend the flow around a cylinder at various velocities and examine the related flow characteristics. The coefficients of drag, pressure, drag force, and Reynolds number are the main parameters that must be examined. Pressure could be measured at different angles thanks to the arrangement, which included a cylinder with a static pressure port in the middle. To investigate the variations in the flow patterns, the coefficients of pressure (C_p), Reynolds number (Re), drag coefficient (C_d), and drag force were computed and compared following the experiment at two distinct speeds. The data showed that the airflow becomes increasingly turbulent and differs from expected flow behavior at specific spots along the cylinder. The fluid flow tends to "stick" to the cylinder and is not separated as rapidly at higher speeds. Subcritical conditions were discovered in both flow regimes. Furthermore, it was shown that the drag coefficient (C_d) and drag force both increased with increasing speed and Reynolds number.

Aim & Objectives

The lab's goal is to comprehend the various parameters of flow around a cylinder.

The objectives were:

- Measurement of surface pressure distributions.
- Examine possible flow solutions in comparison to the experiment.
- Determine a cylinder's drag and coefficient of drag at various Reynolds numbers.
- Analyze uncertainty and the spread of errors.

Uncertainty Analysis

Measurements of the surrounding conditions must be made prior to the experiment, and the computation of any uncertainties. All of the environmental conditions and their uncertainties are listed in the table below:

Quantity	Measurement	Uncertainty
Pressure	102946.2 Pa	16.9 Pa
Temperature	297.15 K	0.5 K
Density	1.207 kg/m ³	N/A

Table 1: Quantity, Measurement, and Uncertainty

The uncertainty of the velocity had to be determined before the uncertainty of the coefficient of drag (C_d) and coefficient of pressure (C_p) could be propagated. This formula is used to determine the velocity's uncertainty:

$$\delta V = \delta V_{P_{dyn}} + \delta V_{P_{atm}} + \delta V_{T_{atm}}$$

In this case, δV represents the velocity uncertainty, $\delta V_{P_{dyn}}$ represents the dynamic pressure uncertainty, $\delta V_{P_{atm}}$ represents the atmospheric pressure uncertainty, and $\delta V_{T_{atm}}$ represents the atmospheric temperature uncertainty. Each of the following formulas is used to determine the other uncertainty values:

$$\delta V_{P_{dyn}} = \sqrt{\frac{2 * (P_{dyn} + \delta P_{dyn})}{\frac{P_{atm}}{R * (T_{atm})}}} - V$$

$$\delta V_{P_{atm}} = \sqrt{\frac{2 * P_{dyn}}{\frac{(P_{atm} + \delta P_{atm})}{R * (T_{atm})}}} - V$$

$$\delta V_{T_{atm}} = \sqrt{\frac{2 * P_{dyn}}{\frac{P_{atm}}{R * (\delta T_{atm})}}} - V$$

Results & Discussion

Two experiments were carried out in this lab. A hole in the middle of a tiny cylinder was used to collect pressure data in both tests, and the cylinder's tilt was changed. A speed of 10 m/s was used for the first trial, and 15 m/s for the second. The static pressure tap's data allows us to compute the drag force (D), coefficient of drag (C_d), and coefficient of pressure (C_p).

The temperature and pressure of the atmosphere must be measured before the experiment is carried out. It was recorded that the pressure was 30.4 inHg and the temperature was 24 °C. This is necessary to determine the air's density. It was also determined that the cylinder's diameter was 1.4 in.

Coefficient of Pressure:

The coefficient of pressure is the initial focus of the analysis. Dynamic pressure is determined by subtracting the ambient pressure from the total pressure before determining the coefficient of pressure. To find the true coefficient of pressure, this information is entered into the equation:

$$C_p(actual) = \frac{P_{dyn}}{\frac{1}{2} * \rho * V^2}$$

The potential C_p and the actual coefficient of pressure will be contrasted. The following is the formula for a cylinder's potential C_p flow:

$$C_p(potential) = 1 - 4\sin^2(\theta)$$

The Reynolds number is calculated:

$$Re = \frac{\rho * V * d}{\mu}$$

Together with the actual measured coefficients of pressure, we may construct a graph showing the potential flow using the computed values.

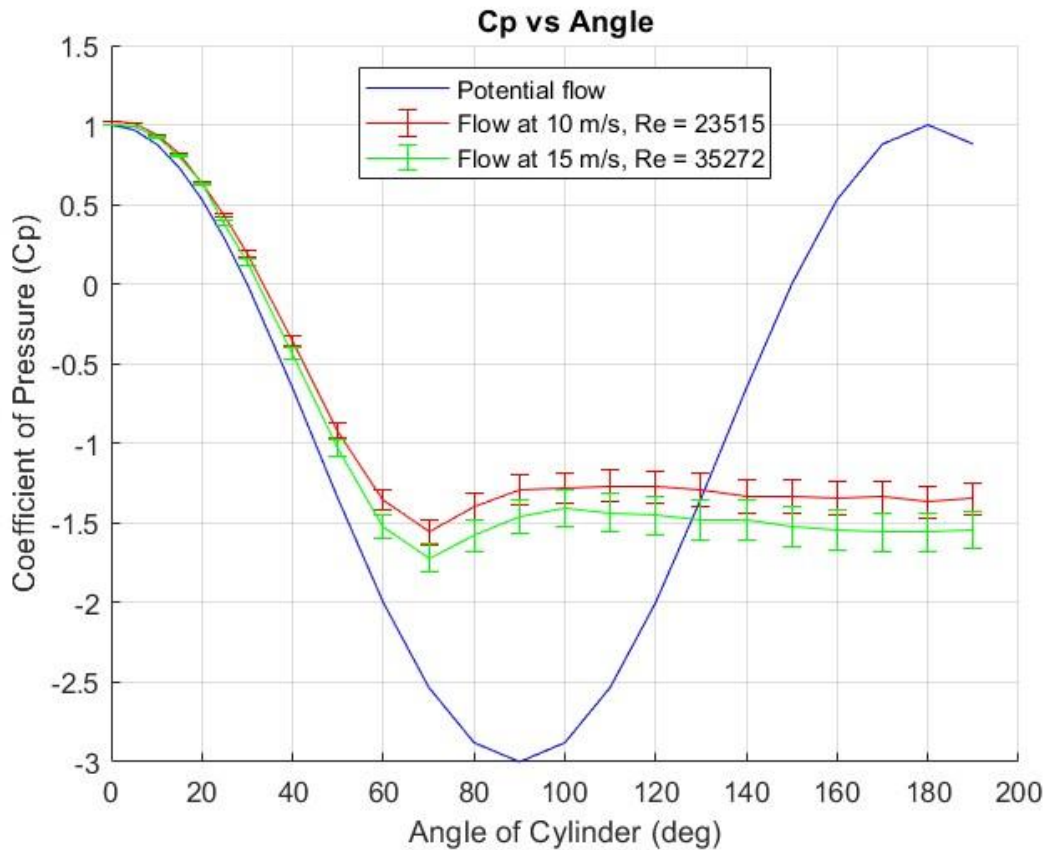


Figure 1: Coefficient of pressure vs Angle of Cylinder

The above figure illustrates how the coefficient of pressure varies with the cylinder's angle. The potential flow is shown by the blue line, and the maximum coefficients of pressure are indicated by the peaks. The red curve represents the flow at 10 m/s and the Reynolds number at this speed is around 23515. The green curve represents the flow at 15 m/s and the Reynolds number at this speed is around 35272. We can see that the flow takes longer to separate from the cylinder when the velocity of the flow is higher. Since both of the Reynolds numbers were less than 300000, it can be concluded that the flows were subcritical.

Coefficient of Drag and Drag Force:

The coefficient of drag (C_d) and Reynold's number are the focus of the second section of the investigation. We may use the following formula to determine the C_d using the C_p determined in the preceding section of the analysis:

$$C_d = \int_0^\pi C_p * \cos\theta d\theta$$

We can determine the C_d for each cylinder angle using MATLAB's "trapz" function, which is plotted against the Reynolds numbers calculated earlier.

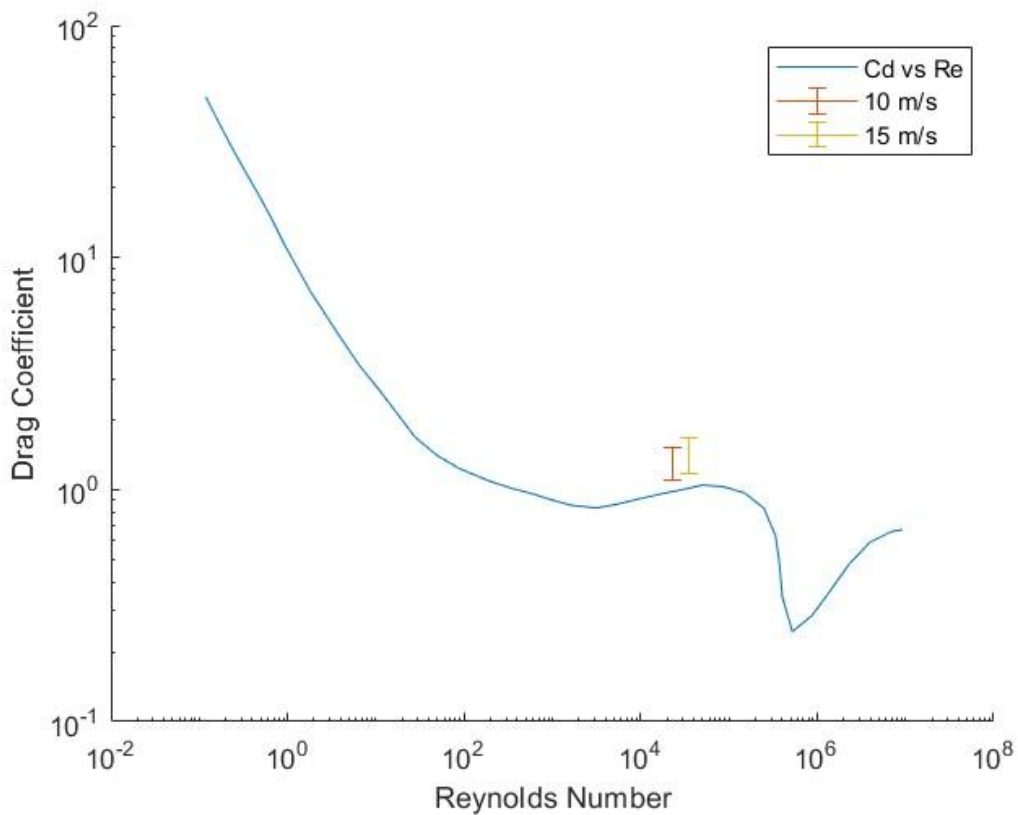


Figure 2: Coefficient of Drag vs Reynolds Number

According to the graph, the calculated Reynolds numbers for each speed were not in line with the theoretical values. Surface roughness, flow separation, and measurement error are a few possible causes of this variation.

Flow Velocity (m/s)	C_d	Uncertainty
10	1.3055	0.0043
15	1.4239	0.0051

Table 2: Uncertainty related to C_d

From the table, it can be interpreted that as the velocity of flow increases, the coefficient of drag also increases, and hence the drag force, since drag is directly proportional to drag coefficient.

Drag Analysis is done using the following equation, where it involves calculating drag force on the cylinder at the two velocities:

$$D = \frac{1}{2} * \rho * V^2 * A * C_d$$

References

1. Experiment CYL – Flow Around Cylinders: (2025). Embry-Riddle Aeronautical University.
2. AE315 Supplement – Experimental Aerodynamics Supplement: (2025). Embry-Riddle Aeronautical University.

Appendix

Appendix A: Density Calculation

$$P = \rho RT$$

$$\rho_{air} = \frac{P}{RT}$$

$$\rho_{air} = \frac{102946.2}{287 * 297.15} = 1.207 \frac{kg}{m^3}$$

Appendix B: Sutherlands Formula

$$\mu = \mu_0 \left(\frac{T}{T_0} \right)^{\frac{3}{2}} * \frac{T_0 + S_\mu}{T + S_\mu}$$

$$(for\ air) \mu_0 = 1.716 * 10^{-5} \frac{N * s}{m^2}, T_0 = 273\ K, S_\mu = 111\ K$$

$$\mu = 1.716 * 10^{-5} \left(\frac{297.15}{273} \right)^{\frac{3}{2}} * \frac{273 + 111}{297.15 + 111} = 1.833 * 10^{-5} \frac{N * s}{m^2}$$

Appendix C: MATLAB Code

```
% Author: Anuranan Bharadwaj
% Lab 02: Experiment CYL
```

```
clc;
clear;
close all;
```

```
%%%%%%%%%%%% ambient conditions %%%%%%%%%%%%%%
```

```
p_atm = 102946.2;
t_atm = 297.15;
R = 287;
density = p_atm/(t_atm*R);
diameter = 0.03556;
```

```
%%%%%%%%%%%% Analysis for 10 m/s %%%%%%%%%%%%%%
```

```
filecount = 24;
```

```
currAngle_10ms = zeros(1, filecount);
dp_10ms = zeros(1, filecount);
sd_10ms = zeros(1, filecount);
v_10ms = zeros(1, filecount);
```

```
for i = 1:filecount
    file_10ms = load(sprintf('LAB2_DATA/10ms/Experiment2_10ms_%d.mat', i));
```

```
    dp_10ms(i) = mean(file_10ms.dp)*248.84;
    currAngle_10ms(i) = file_10ms.currAngle;
    sd_10ms(i) = std(dp_10ms);
    v_10ms(i) = sqrt((2 * dp_10ms(i))/density);
```

```
    v_1 = 10;
    Cp_10ms(i) = (dp_10ms(i))/(.5 * density * v_1^2) + 1;
end
```

```
%%%%%%%%%%%% Analysis for 15 m/s %%%%%%%%%%%%%%
```

```
currAngle_15ms = zeros(1, filecount);
dp_15ms = zeros(1, filecount);
sd_15ms = zeros(1, filecount);
v_15ms = zeros(1, filecount);
```

```
for i = 1:filecount
    file_15ms = load(sprintf('LAB2_DATA/15ms/Experiment2_15ms_%d.mat', i));
```

```
    dp_15ms(i) = mean(file_15ms.dp)*248.84;
    currAngle_15ms(i) = file_15ms.currAngle;
    sd_15ms(i) = std(dp_15ms);
```

```

v_15ms(i) = sqrt((2 * dp_15ms(i))/density);

v_2 = 15;
Cp_15ms(i) = (dp_15ms(i))/(.5 * density * v_2^2) + 1;
end

% Cp - potential
Cp_potential = 1 - 4.*sind(currAngle_10ms(2:24)).^2;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Uncertainty - Cp %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
dp_uc_10ms = (.007877 * sd_10ms) / sqrt(filecount);
dp_uc_15ms = (.003877 * sd_15ms) / sqrt(filecount);

delP = 0.01*248.8;
dAngle = 0.5;
dT = 0.5;
dP = 16.9;

dV_10ms = sqrt(((1 / (2 * sqrt(2 * max(dp_10ms) / density))) * dp_uc_10ms(1))^2 + ...
    ((-(max(dp_10ms)) / (2 * density^2 * sqrt(2 * max(dp_10ms)/ density))) * ...
    (dP / (R*t_atm)))^2);

dV_15ms = sqrt(((1 / (2 * sqrt(2 * max(dp_15ms) / density))) * dp_uc_15ms(1))^2 + ...
    ((-(max(dp_15ms)) / (2 * density^2 * sqrt(2 * max(dp_15ms)/ density))) * ...
    (dP / (R*t_atm)))^2);

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Drag Force %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
theta1 = deg2rad(currAngle_10ms);
theta = deg2rad(currAngle_10ms(2:23));
freq_10 = (Cp_10ms(2:23)) .* cos(theta);
freq_15 = (Cp_15ms(2:23)) .* cos(theta);
Cd_10 = trapz(theta, freq_10);
Cd_15 = trapz(theta, freq_15);

DF_10 = 0.5 * Cd_10 * density * diameter * 100;
DF_15 = 0.5 * Cd_15 * density * diameter * 15^2;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Uncertainty - Drag %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
Cd_uc_10 = sqrt(sum((cos(theta) .* dp_uc_10ms(2:23)).^2) / filecount) * (theta1(2) - ↵
theta1(1));
Cd_uc_15 = sqrt(sum((cos(theta) .* dp_uc_15ms(2:23)).^2) / filecount) * (theta1(2) - ↵
theta1(1));

% Reynolds Number

```

```
mu = 1.833*10^(-5); %N*s/m2
Re_10 = (density * 10 * diameter) / mu;
Re_15 = (density * 15 * diameter) / mu;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% Plotting %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Cp vs Angle
figure(1)
hold on

plot(currAngle_10ms(2:24), Cp_potential, 'b') %potential curve
errorbar(currAngle_10ms(2:24), Cp_10ms(2:24), dp_uc_10ms(2:24), 'r'); %10 m/s curve
errorbar(currAngle_15ms(2:24), Cp_15ms(2:24), dp_uc_15ms(2:24), 'g'); %15 m/s curve

xlabel('Angle of Cylinder (deg)');
ylabel('Coefficient of Pressure (Cp)');
legend('Potential flow', 'Flow at 10 m/s, Re = 23515', 'Flow at 15 m/s, Re = 35272', 'Location', 'north');
title('Cp vs Angle');

grid on
hold off

% Cd vs Re
Cd_line = readtable("LAB2_DATA\CYL_Data.csv");

figure(2)
hold on
plot(Cd_line.Var1, Cd_line.Var2);
set(gca, 'YScale', 'log', 'XScale', 'log')
errorbar(Re_10, Cd_10, Cd_uc_10*50)
errorbar(Re_15, Cd_15, Cd_uc_15*50)
xlabel('Reynolds Number')
ylabel('Drag Coefficient')
legend('Cd vs Re', sprintf('10 m/s'), sprintf('15 m/s'))
hold off
```